Camunda Cancel Event

In this session we will look at what is cancel event and how it functions ?.

Cancel event is used to cancel the transaction and to trigger compensation.

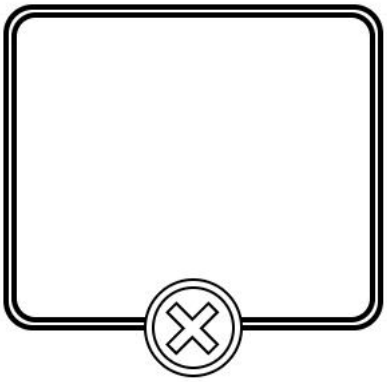
Types of compensation events

1. End cancel event
2. Intermediate boundary cancel event

Cancel end event

The cancel end event can only be used in combination with a transaction subprocess. When the cancel end event is reached, a cancel event is thrown which must be caught by a cancel boundary event. The cancel boundary event then cancels the transaction and triggers compensation.

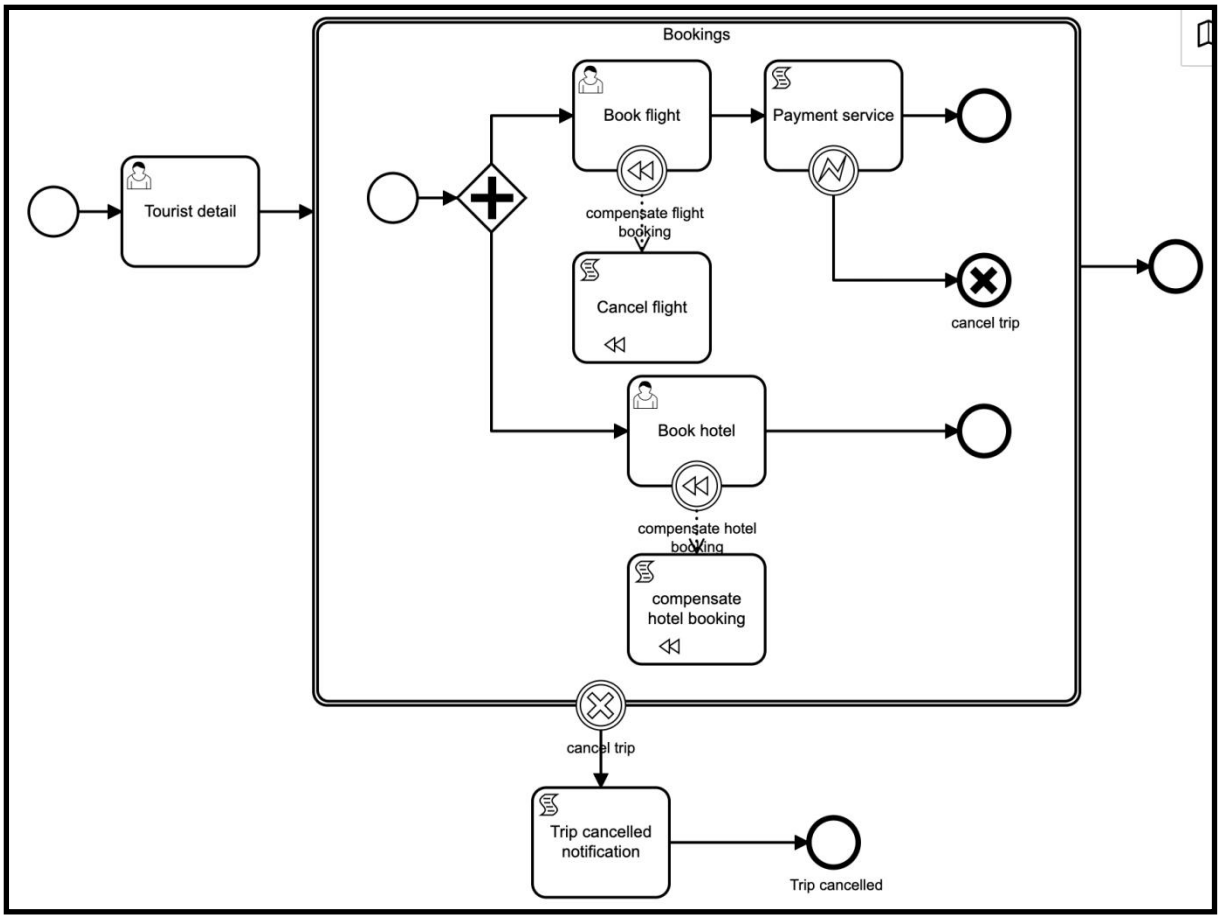
Cancel boundary event



An attached intermediate catching cancel event on the boundary of a transaction subprocess is triggered when a transaction is canceled. When the cancel boundary event is triggered, it first interrupts all executions active in the current scope. Next, it starts compensation of all active compensation boundary events in the scope of the transaction.

Compensation is performed synchronously, i.e. the boundary event waits before compensation is completed before leaving the transaction. When compensation is completed, the transaction subprocess is left using the sequence flow(s) running out of the cancel boundary event.

1. To demonstrate the use of cancel event we will use following process diagram.

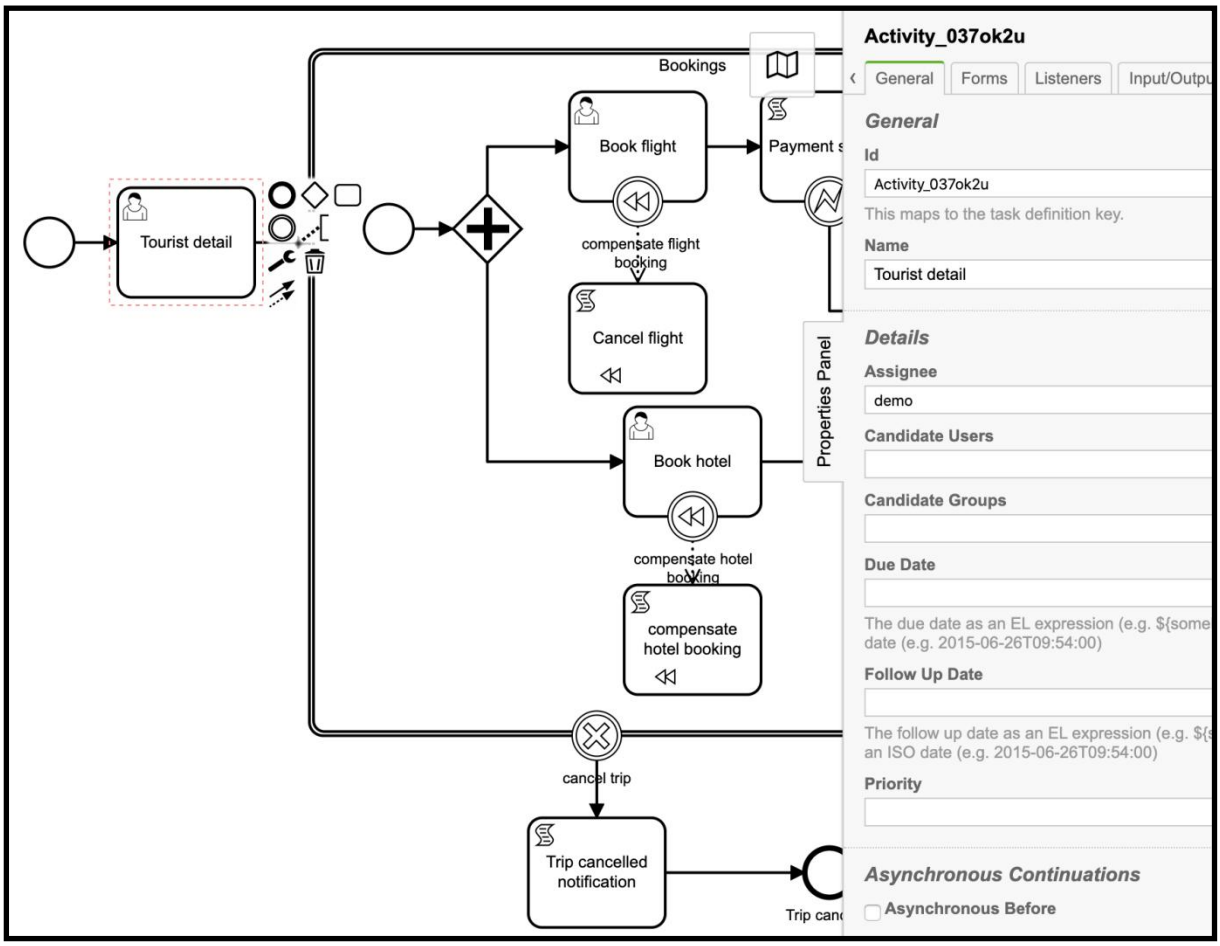


Properties:

Id: Process_Cancel_Event

Name: Process cancel event

2. User task: Tourist detail

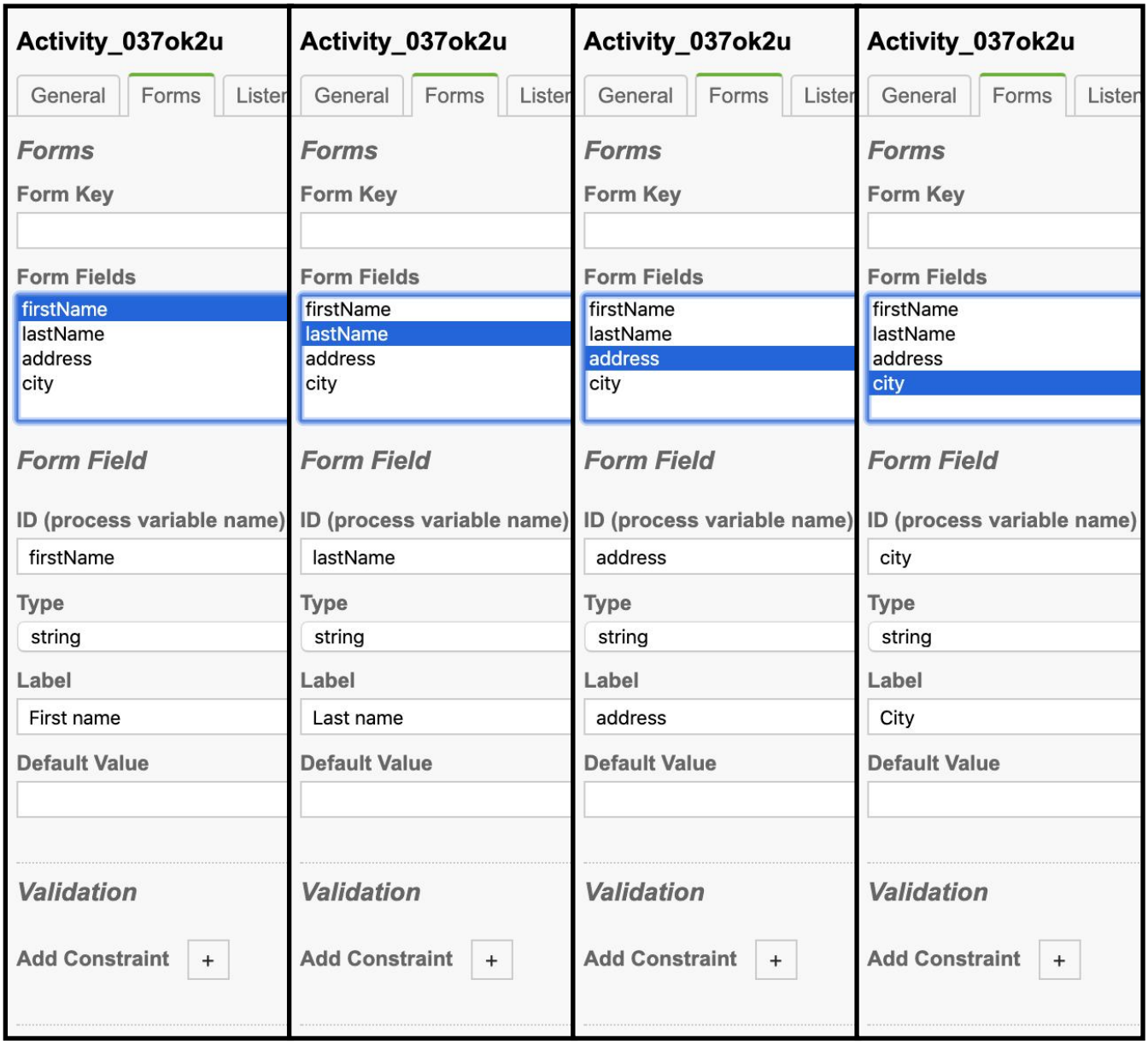


Id: Activity_037ok2u

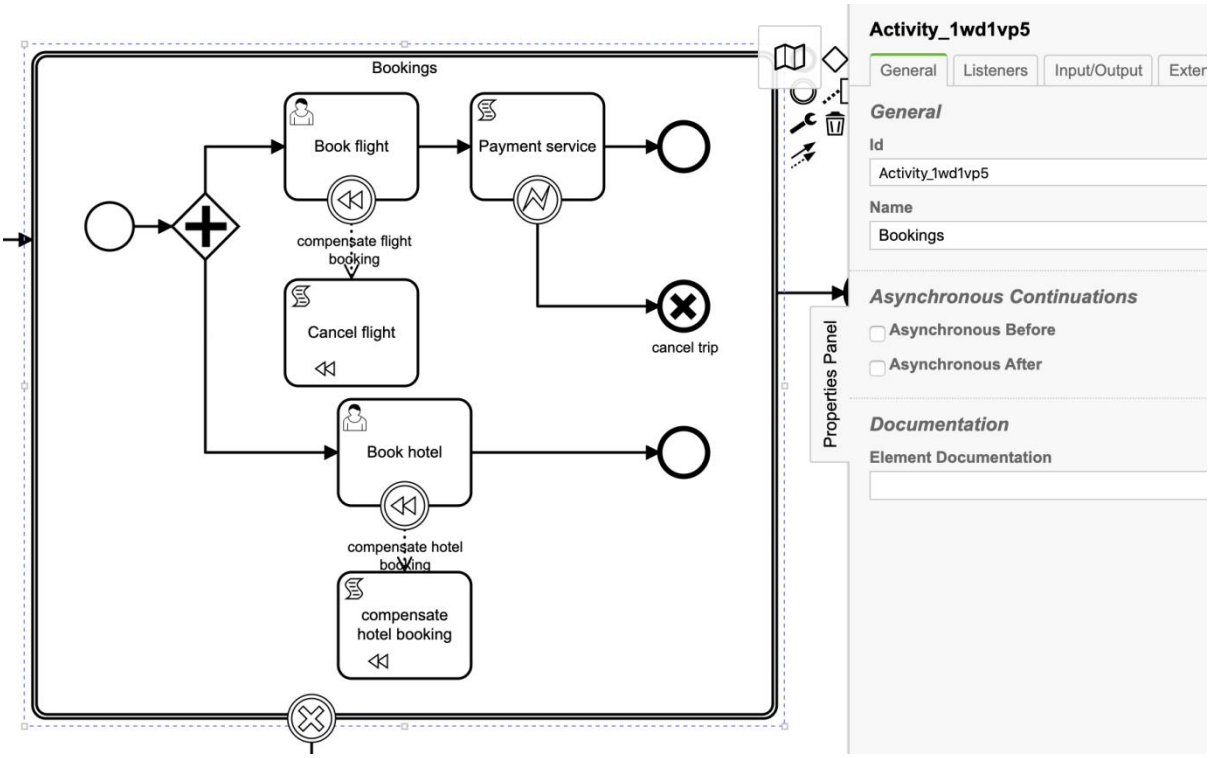
Name: Tourist detail

Assignee: demo

Task form:


Activity_037ok2u
General
Forms
Listen
Forms
Form Key
Form Fields
firstName
lastName
address
city
Form Field
ID (process variable name)
firstName
Type
string
Label
First name
Default Value
Validation
Add Constraint
+
Activity_037ok2u
General
Forms
Listen
Forms
Form Key
Form Fields
firstName
lastName
address
city
Form Field
ID (process variable name)
lastName
Type
string
Label
Last name
Default Value
Validation
Add Constraint
+
Activity_037ok2u
General
Forms
Listen
Forms
Form Key
Form Fields
firstName
lastName
address
city
Form Field
ID (process variable name)
address
Type
string
Label
address
Default Value
Validation
Add Constraint
+
Activity_037ok2u
General
Forms
Listen
Forms
Form Key
Form Fields
firstName
lastName
address
city
Form Field
ID (process variable name)
city
Type
string
Label
City
Default Value
Validation
Add Constraint
+

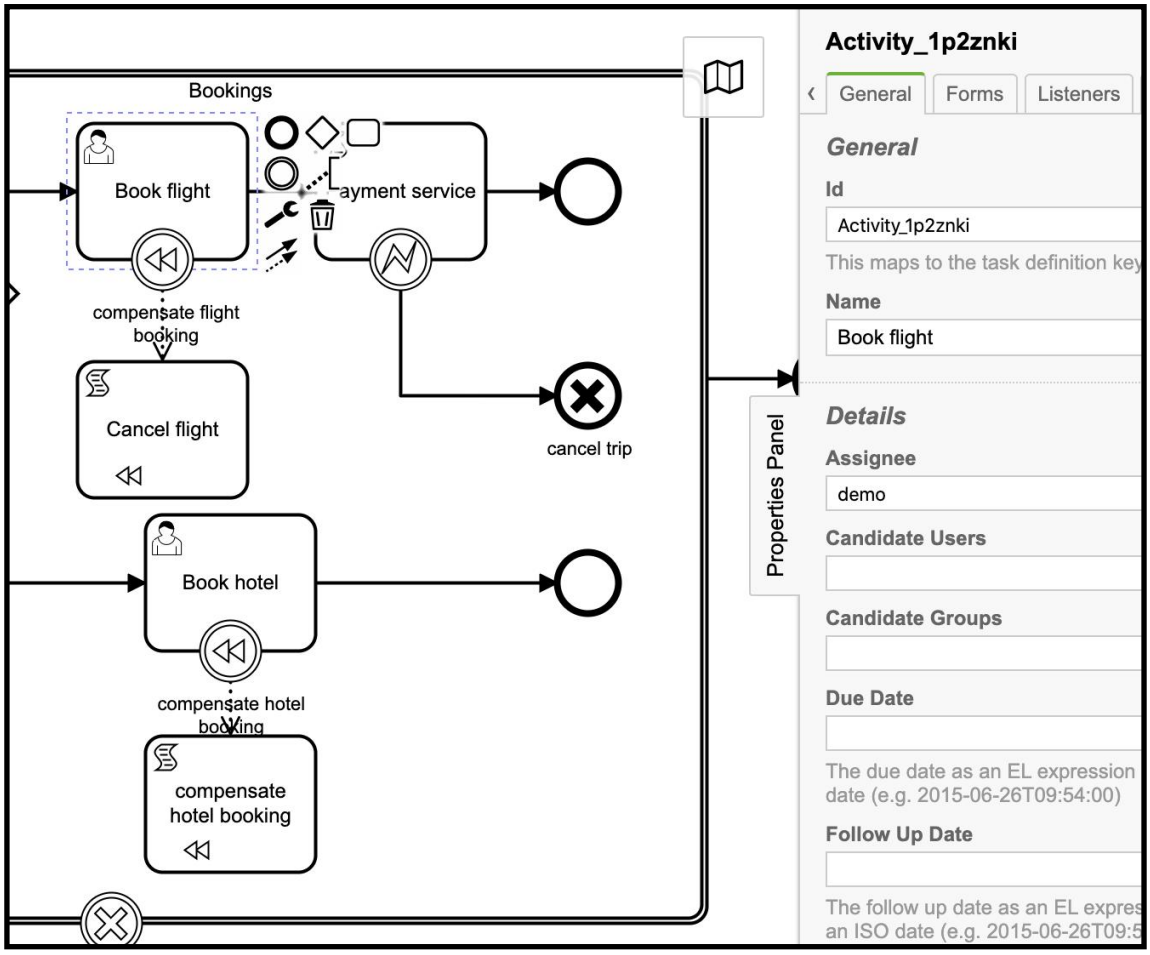
3. Transactional sub process: Bookings



Id: Activity_1wd1vp5

Name: Bookings

4. User task: Book flight

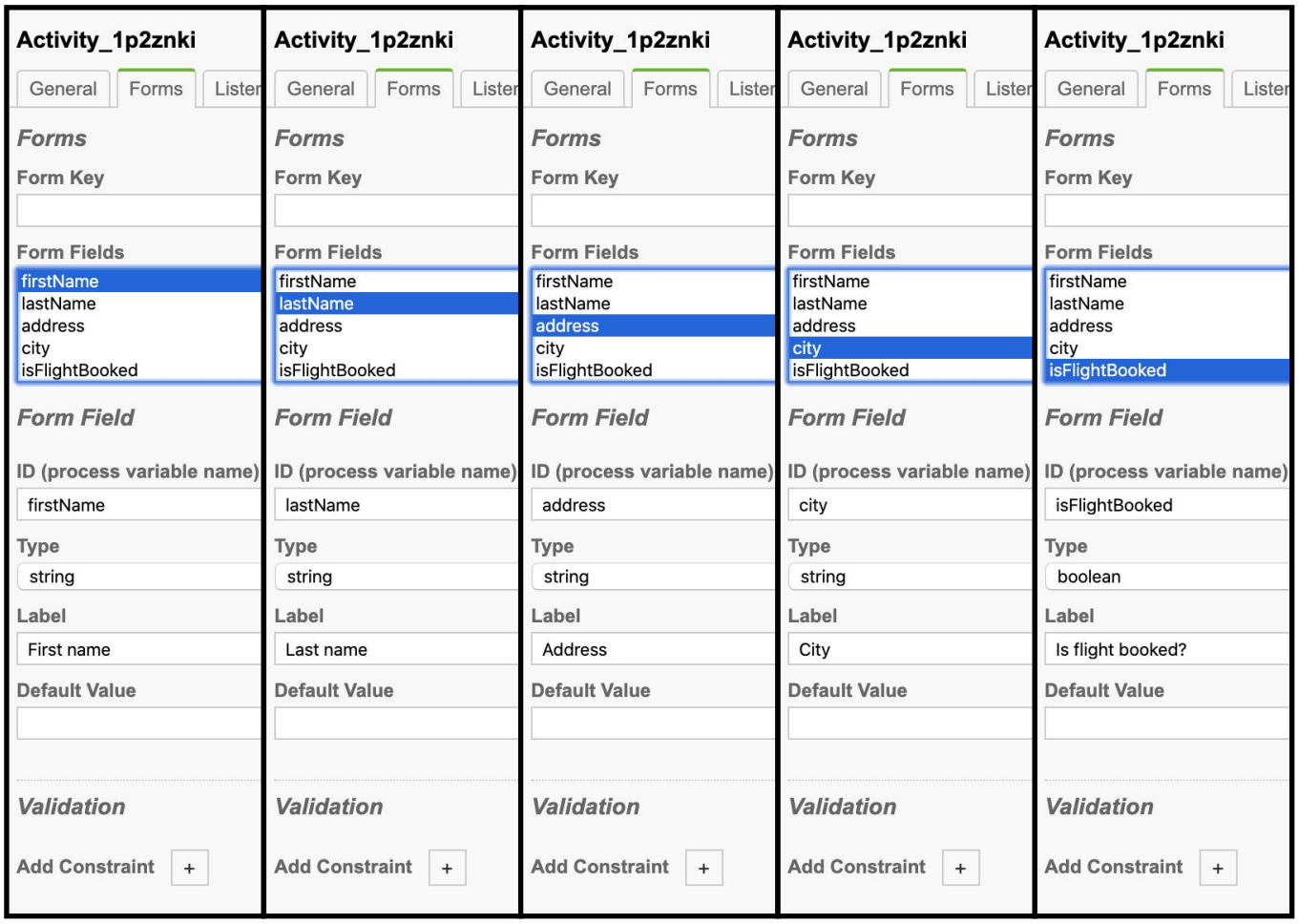


Id: Activity_1p2znki

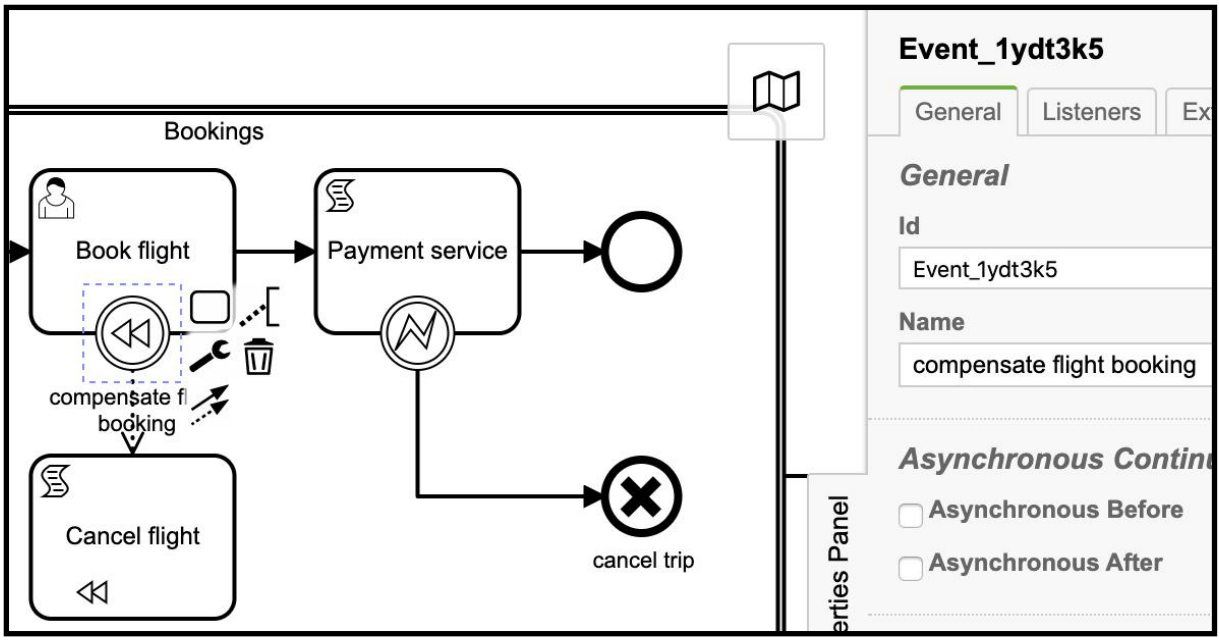
Name: Book flight

Assignee: demo

Task form:



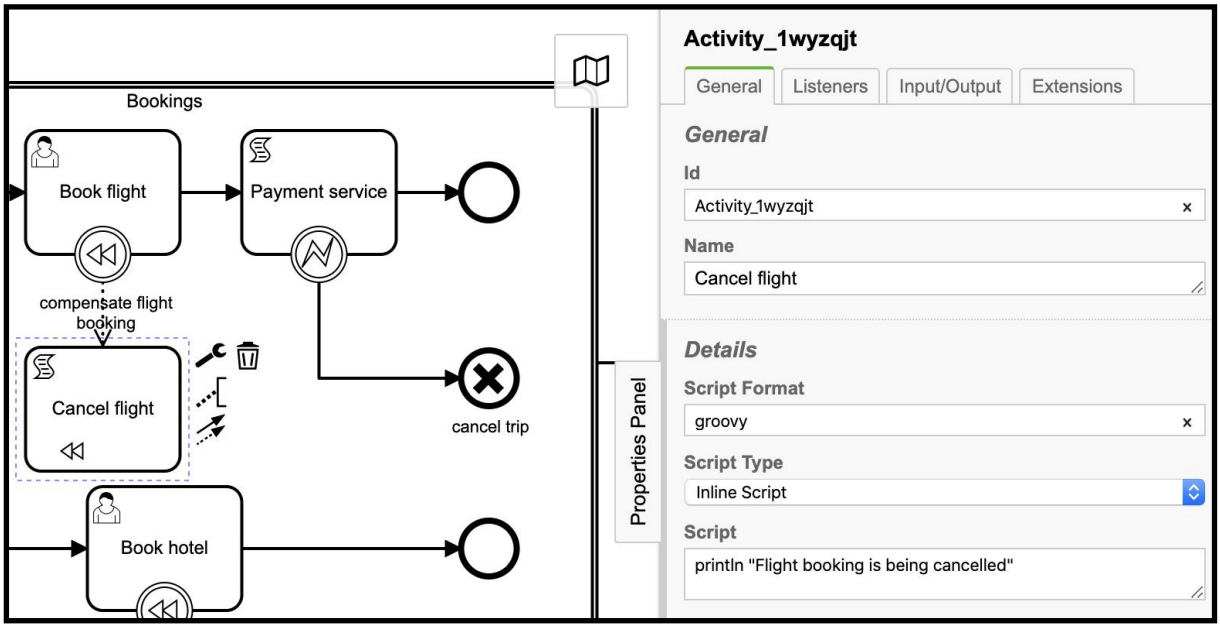
5. Boundary compensate event



Id: Event_1ydt3k5

Name: compensate flight booking

6. Script task: Cancel flight



Id: Activity_1wyzqjt

Name: Cancel flight

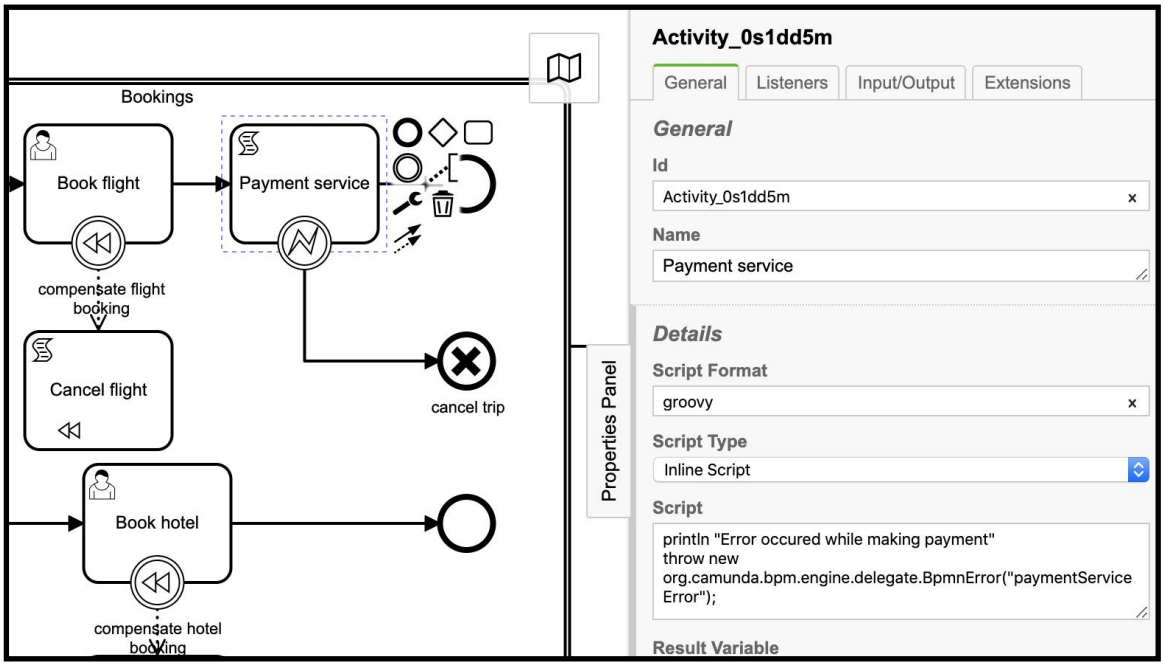
Script Format: groovy

Script Type: Inline Script

Script:

```
println "Flight booking is being cancelled"
```

7. Script task: Payment service



Id: Activity_0s1dd5m

Name: Payment service

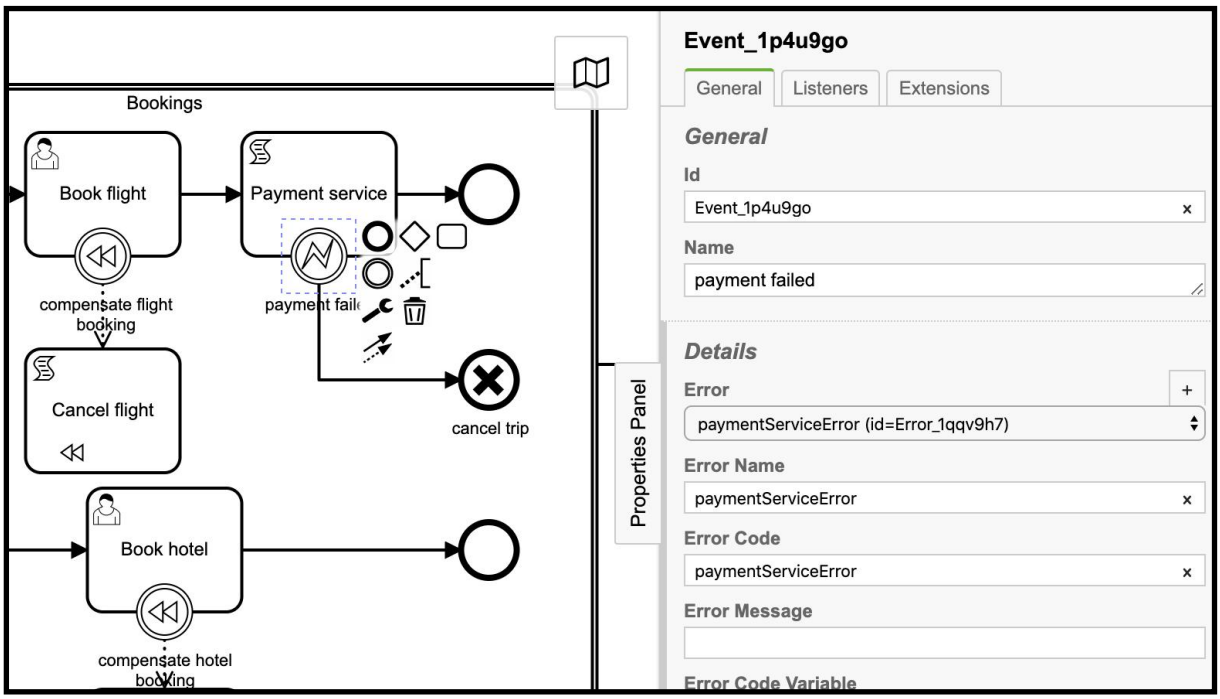
Script Format: groovy

Script Type: Inline Script

Script:

```
println "Error occured while making payment"
throw new
org.camunda.bpm.engine.delegate.BpmnError("paymentServiceError");
```

8. Boundary error event: Payment failed



Id: Event_1p4u9go

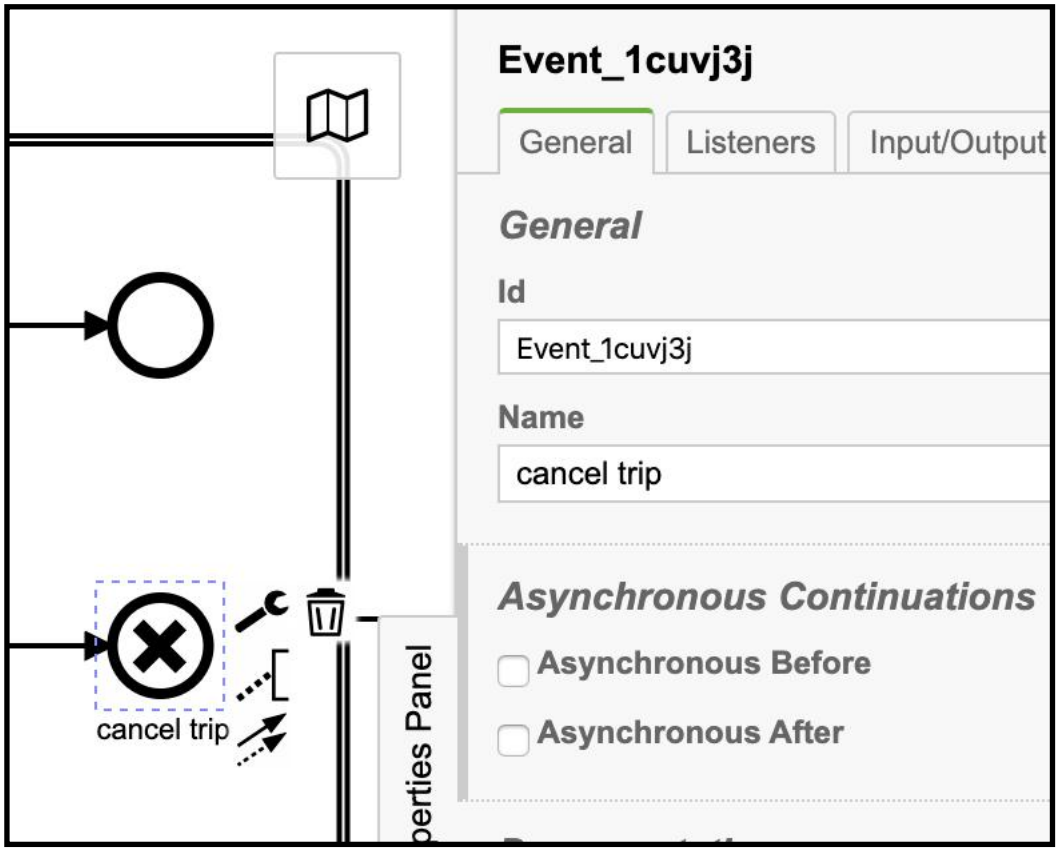
Name: payment failed

Error: paymentServiceError (id=Error_1qqv9h7)

Error Name: paymentServiceError

Error Code: paymentServiceError

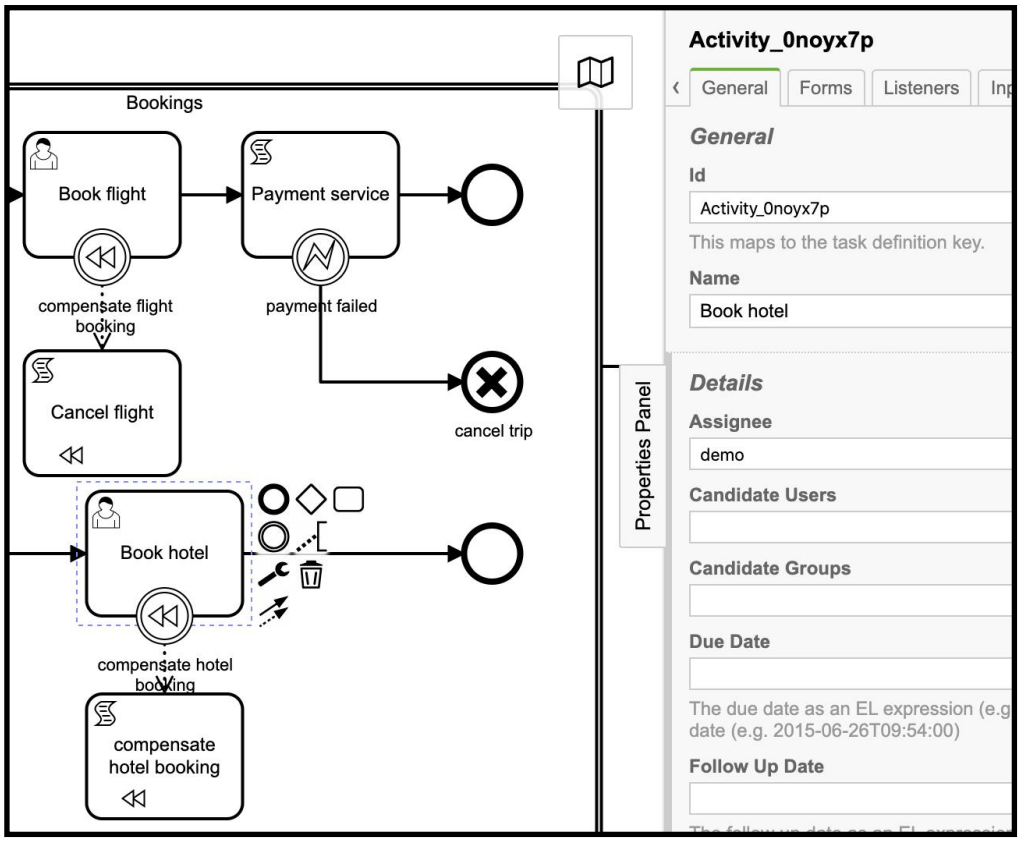
9. Cancel end event: cancel trip



Id: Event_1cuvj3j

Name: cancel trip

10. User task: Book hotel

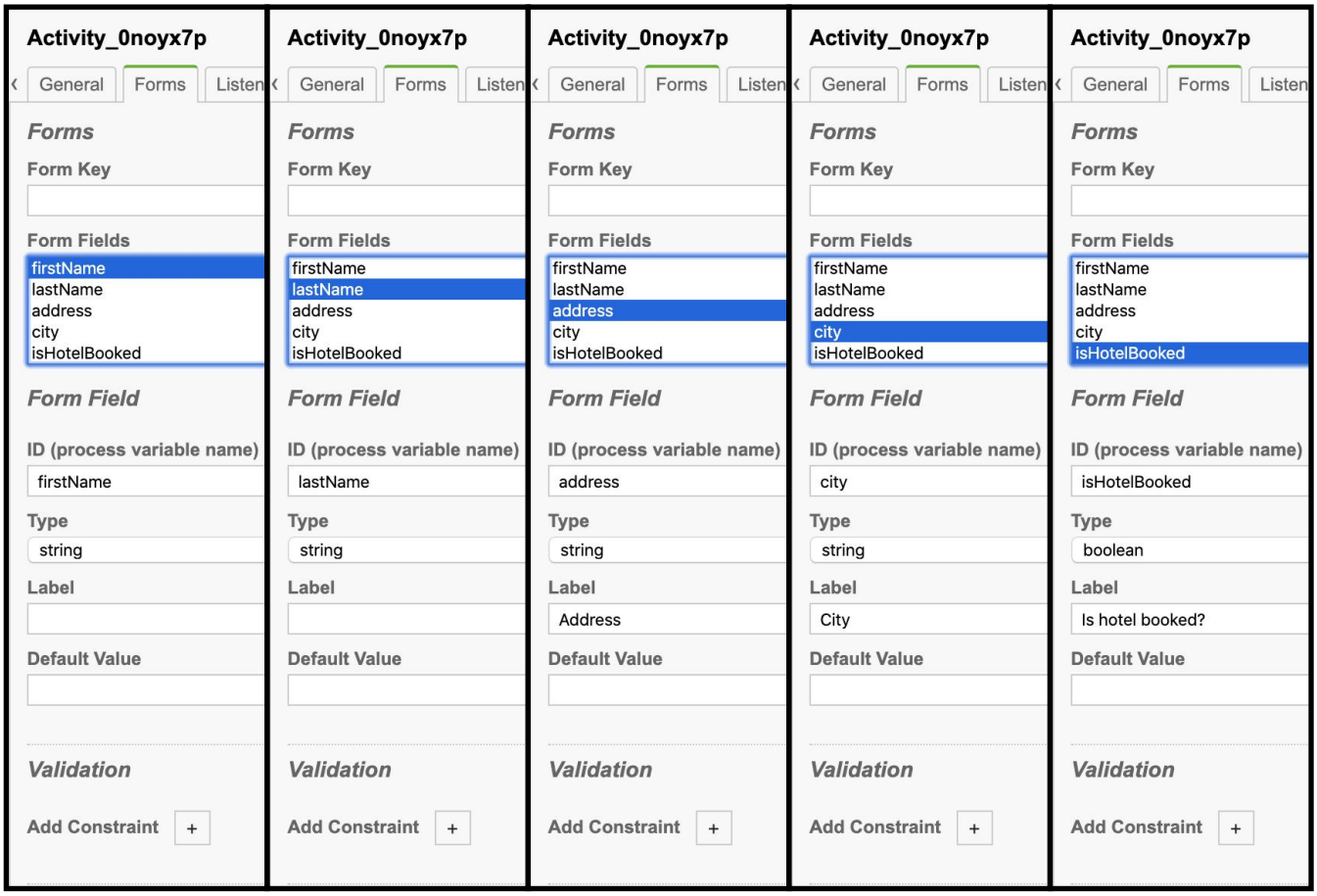


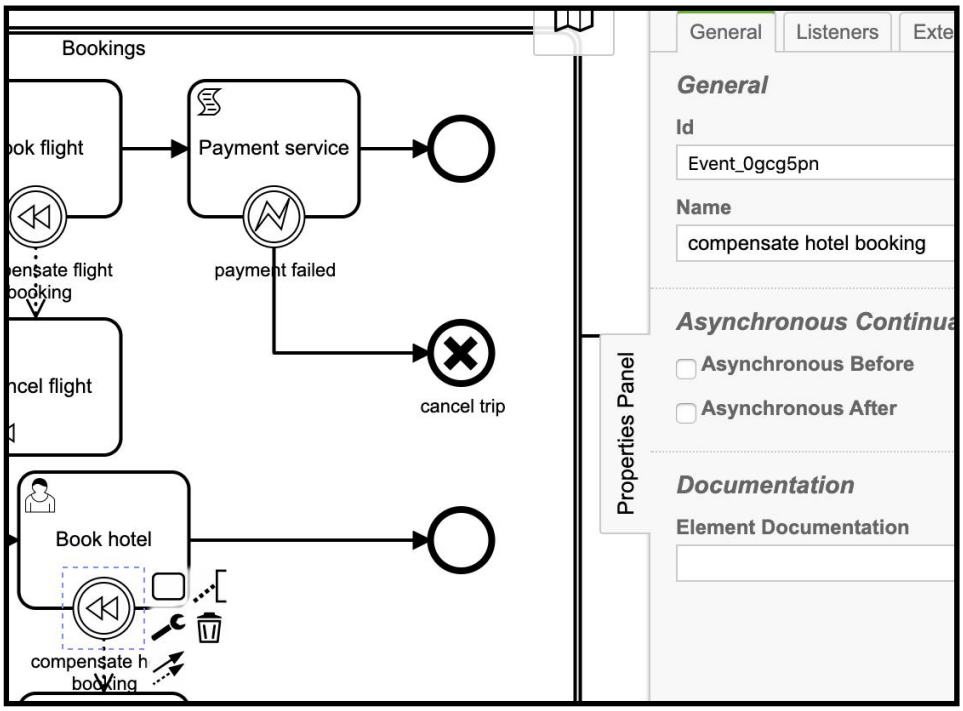
Id: Activity_0noyx7p

Name: Book hotel

Assignee: demo

Task form:



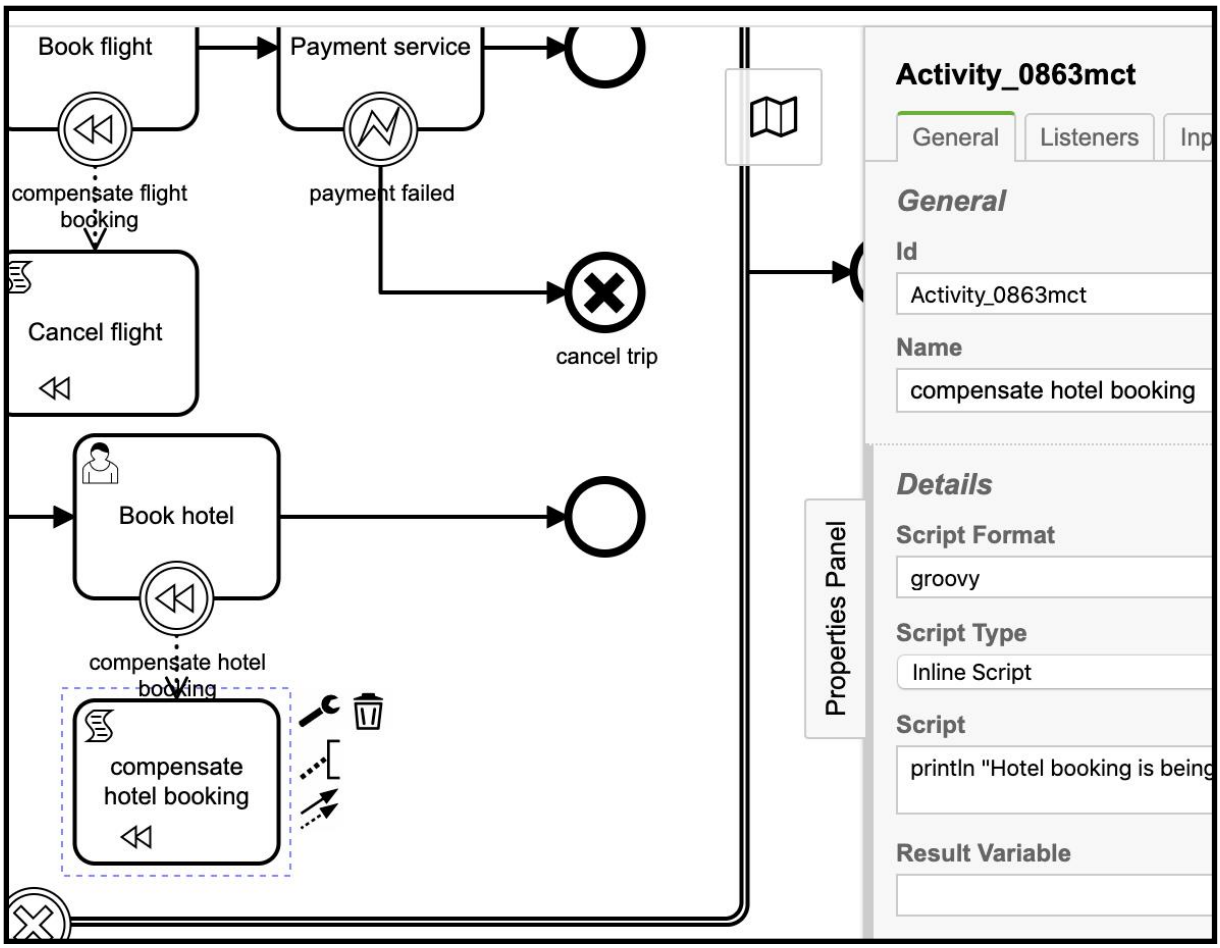


11. Boundary compensate event: compensate hotel booking

Id: Event_0gcg5pn

Name: compensate hotel booking

11. Script task: compensate hotel booking



Id: Activity_0863mct

Name: compensate hotel booking

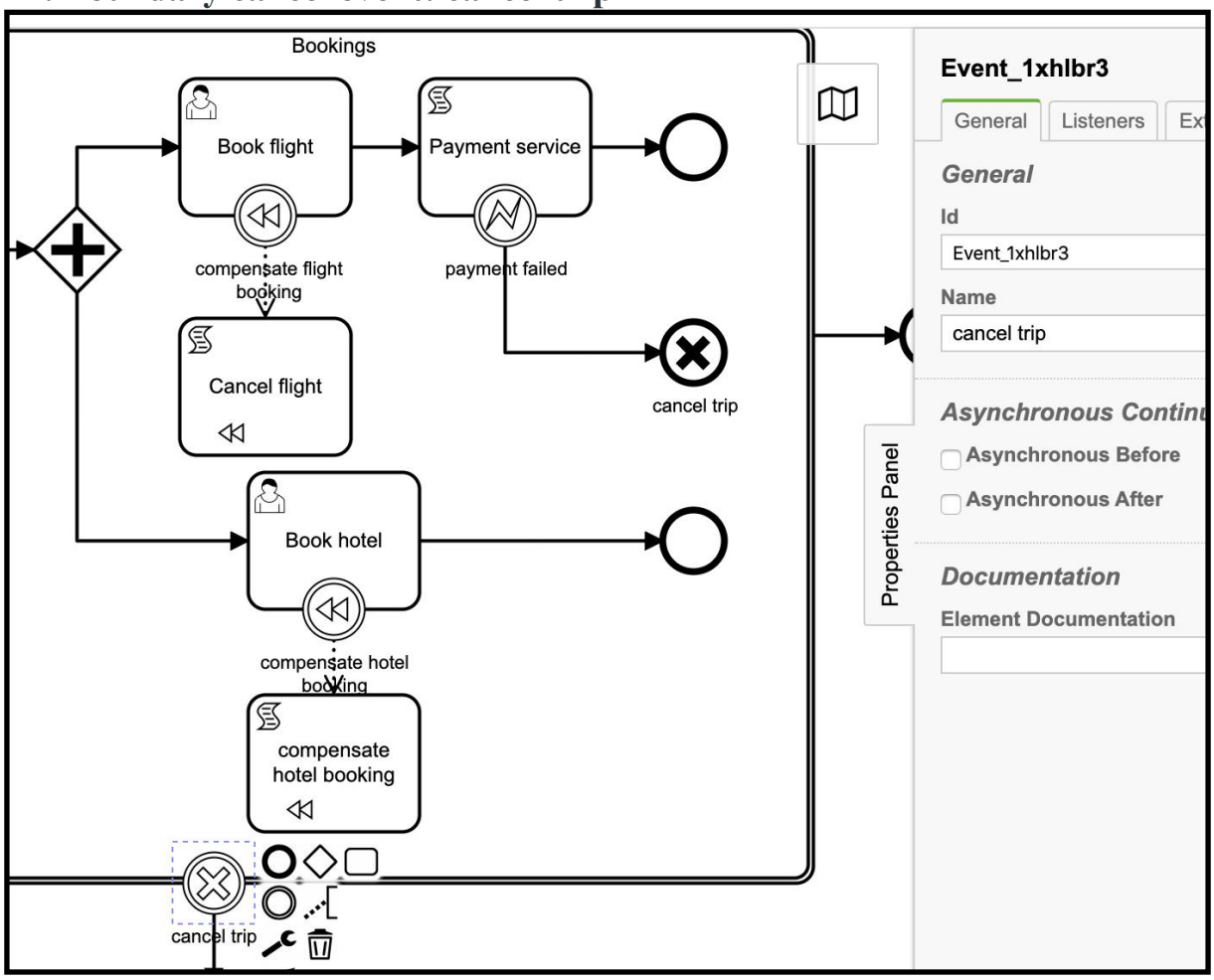
Script Format: groovy

Script Type: Inline Script

Script:

```
println "Hotel booking is being cancelled"
```

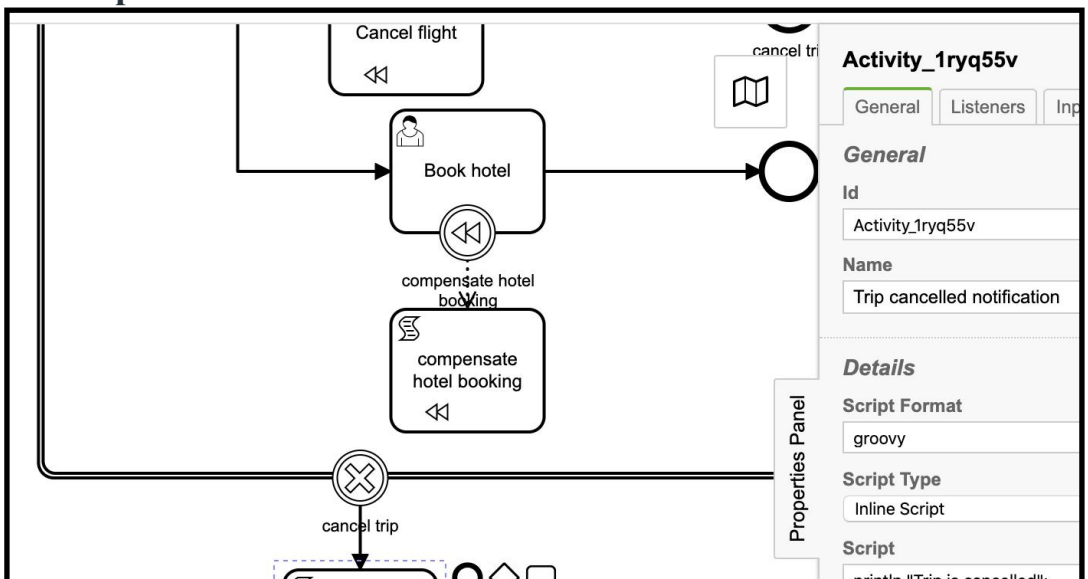
12. Boundary cancel event: cancel trip



Id: Event_1xhlbr3

Name: cancel trip

13. Trip cancelled notification



Id: Activity_1ryq55v

Name: Trip cancelled notification**Script Format:** groovy

Script Type: Inline Script

Script:

```
        println "Trip is cancelled”;
```

14. Deploy and start the process

Hit the process cancel event and hit the Start button to start the process.

User task Tourist detail will get created as per process definition.

Fill-up the user task form and hit the Complete button.

As per process definition, two user tasks Book hotel and Book flight user task will get created simultaneously.

Lets first complete the user task Book hotel

Fill-up the user task form and hit the Complete button.

Similarly, complete the user task Book flight

Fill-up the user task form and hit the Complete button.

As per process definition, after completion of user task **Book flight,** script task **payment service** will execute and it will throw error which will be caught by boundary error **event payment failed** then end cancel event will get triggered which will be caught by boundary cancel event **cancel trip** and on execution of boundary cancel event it will execute all the compensate event inside the sub process to compensate the user tasks.

Observe the tomcat console logs

Error occured while making payment
Flight booking is being cancelled
Hotel booking is being cancelled
Trip is cancelled